



IDX G9 CHEMISTRY S+ STUDY GUIDE

ISSUE S2 FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

I. Metallic Bonding and Properties of Metals

1. Atomic Structure of Metals

- **Key Components:**
 - **Cations:** Positively charged metal ions formed by losing valence electrons (e.g., Na^+ , Mg^{2+}).
 - **Electron Sea:** Delocalized valence electrons freely moving throughout the metal lattice.
- **Model Analogy:** Think of metal cations as "islands" submerged in a "sea" of electrons that hold them together.

2. Physical Properties Explained by Metallic Bonding

Property	Explanation
Metallic Luster	Free electrons absorb and re-emit light across the visible spectrum.
Conductivity	Mobile electrons carry electric current (charge) and thermal energy (vibrations).

Property	Explanation
Malleability	Applying force causes cations to slide, but electron sea prevents repulsion and 断裂 (breaking).
Ductility	Cations can be drawn into wires without breaking bonds, supported by electron sea.
High Density	Close-packed crystal structures (e.g., face-centered cubic) in most metals.

3. Strength of Metallic Bonding

- **Primary Factors:**
 1. **Ionic Radius:**
 - **Trend:** Smaller radius = stronger attraction to electron sea (e.g., $\text{Li}^+ < \text{Na}^+ < \text{K}^+$).
 - **Example:** Melting point of Li (180°C) $>$ Na (98°C) $>$ K (63°C).
 2. **Number of Valence Electrons:**
 - **Trend:** More electrons = stronger bonding (e.g., Na [$1e^-$] $<$ Mg [$2e^-$] $<$ Al [$3e^-$]).
 - **Example:** Melting point of Al (660°C) $>$ Mg (650°C) $>$ Na (98°C).
- **Secondary Factor:** Crystal structure (e.g., close-packed structures enhance bonding).

4. Metallic vs. Ionic Bonds: Key Contrasts

Feature	Metallic Bond	Ionic Bond
Bonding Particles	Cations + electron sea	Cations + anions (electrostatic attraction)
Electron Movement	Delocalized (free to move)	Localized (fixed in lattice)
Conductivity (Solid)	High (electrons mobile)	Low (ions fixed in place)
Mechanical Behavior	Malleable/ductile (flexible)	Brittle (shatters under force)
Examples	Fe, Cu, Al	NaCl, MgO

5. Real-World Applications

- **Alloys:** Mixtures of metals (e.g., steel = $\text{Fe} + \text{C}$) enhance strength by disrupting cation packing.

- **Thermal Conductors:** Copper (excellent conductivity) used in wiring and heat sinks.

II. Molecular Geometry: VSEPR Theory and Applications

1. Core Principles of VSEPR Theory

- **Aim:** Predict 3D molecular shapes by minimizing repulsion between **electron domains** (bonding and nonbonding pairs).
- **Key Rules:**
 1. **Lone Pair Repulsion:** Lone pairs repel more strongly than bonding pairs (e.g., H₂O's bent shape due to 2 lone pairs).
 2. **Double/Triple Bonds:** Count as 1 electron domain (e.g., CO₂'s double bonds = 2 domains → linear).
 3. **Bond Angle Trends:**
 - Lone pairs compress bond angles (e.g., NH₃: 107° vs. CH₄: 109.5°).
 - Multiple bonds exert slightly more repulsion than single bonds.

2. Step-by-Step Shape Prediction

1. **Draw the Lewis Structure:**
 - Identify the central atom and count valence electrons.
 - Example: NH₃ has 5 (N) + 3(H) = 8 electrons → 3 bonding pairs + 1 lone pair.
2. **Count Electron Domains:**
 - Bonding domains = number of bonds (single/double/triple = 1 domain each).
 - Nonbonding domains = number of lone pairs.
 - Example: NH₃ has 4 domains (3 bonding + 1 nonbonding).
3. **Determine Electron Domain Geometry:**
 - 2 domains = linear, 3 = trigonal planar, 4 = tetrahedral, etc.
4. **Predict Molecular Shape:**
 - Ignore lone pairs for the final shape (e.g., 4 domains with 1 lone pair = trigonal pyramidal).

3. Detailed Table of Electron Domains and Shapes

# Domains	Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Shape	Bond Angle	Example
2	Linear	2	0	Linear	180°	CO ₂ , HCN
3	Trigonal planar	3	0	Trigonal planar	120°	BF ₃ , SO ₃
3	Trigonal planar	2	1	Bent (V-shaped)	~117°	SO ₂ , NO ₂ ⁻

# Domains	Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Shape	Bond Angle	Example
4	Tetrahedral	4	0	Tetrahedral	109.5°	CH ₄ , CCl ₄
4	Tetrahedral	3	1	Trigonal pyramidal	107°	NH ₃ , PCl ₃
4	Tetrahedral	2	2	Bent	104.5°	H ₂ O, SCl ₂
5	Trigonal bipyramidal	5	0	Trigonal bipyramidal	90°/120°	PF ₅ , PCl ₅
5	Trigonal bipyramidal	4	1	Seesaw	~90°/120°	SF ₄ , TeCl ₄
5	Trigonal bipyramidal	3	2	T-shaped	~90°	ClF ₃ , BrF ₃
5	Trigonal bipyramidal	2	3	Linear	180°	XeF ₂ , I ₃ ⁻
6	Octahedral	6	0	Octahedral	90°	SF ₆ , MoF ₆
6	Octahedral	5	1	Square pyramidal	~90°	BrF ₅ , IF ₅
6	Octahedral	4	2	Square planar	90°	XeF ₄ , PtCl ₄ ²⁻

4. Molecular Polarity: Key Rules

- **Nonpolar Molecules:**
 1. Symmetrical shape (linear, trigonal planar, tetrahedral, etc.).
 2. All surrounding atoms identical (e.g., CO₂, CCl₄, BF₃).
- **Polar Molecules:**

1. Asymmetrical shape (bent, trigonal pyramidal, seesaw, etc.).
 2. Surrounding atoms differ in electronegativity (e.g., H₂O, NH₃, CH₂Cl₂).
- **Example Analysis:**
 - CH₄: Nonpolar (tetrahedral, all H atoms).
 - CH₃Cl: Polar (tetrahedral, Cl is more electronegative than H).

III. Hybridization and Bonding Theories

1. Hybrid Orbitals: Formation and Types

- **Definition:** Mixing of atomic orbitals (s, p, d) to form new hybrid orbitals with equal energy and specific geometries.
- **Key Principle:** Number of hybrid orbitals = number of mixed atomic orbitals.

Hybridization	Atomic Orbitals Mixed	Electron Domains	Geometry	Bond Angle	Examples
sp	1 s + 1 p	2	Linear	180°	BeF ₂ , C ₂ H ₂ (acetylene)
sp ²	1 s + 2 p	3	Trigonal planar	120°	BF ₃ , C ₂ H ₄ (ethylene)
sp ³	1 s + 3 p	4	Tetrahedral	109.5°	CH ₄ , NH ₃ , H ₂ O
sp ³ d	1 s + 3 p + 1 d	5	Trigonal bipyramidal	90°/120°	PF ₅ , SF ₄ , BrF ₃
sp ³ d ²	1 s + 3 p + 2 d	6	Octahedral	90°	SF ₆ , XeF ₄ , IF ₅

2. Sigma (σ) and Pi (π) Bonds

Bond Type	Formation	Strength	Rotation	Occurrence
σ (Sigma)	Head-on overlap of orbitals (s-s, s-p, p-p)	Strongest	Free	All single bonds; 1 per multiple bond
π (Pi)	Side-by-side overlap of parallel p orbitals	Weaker than σ	Restricted	Double bonds (1 π), triple bonds (2 π)

- **Example:**
 - **C₂H₄ (Ethylene):**
 - 1 C=C double bond = 1 σ (sp²-sp² overlap) + 1 π (p-p side overlap).
 - Remaining σ bonds: C-H (sp²-s overlap).
 - **C₂H₂ (Acetylene):**
 - 1 C $\equiv$ C triple bond = 1 σ (sp-sp overlap) + 2 π (p-p side overlaps).

3. Delocalization and Resonance

- **Resonance Structures:** Multiple valid Lewis structures for a molecule (e.g., NO₃⁻, O₃).
- **Delocalization:** π electrons are spread over three or more atoms, stabilizing the molecule.
 - **Example:** NO₃⁻ has 3 resonance structures; its π electrons are delocalized over the N-O bonds.
- **Effect:** Lower energy (more stable) than any single resonance structure.

IV. Practice Problems with Solutions

1. Metallic Bonding Questions

- **Q:** Why is Mg's melting point (650°C) higher than Na's (98°C)?
 - **A:** Mg²⁺ has a smaller radius and more valence electrons (2e⁻ vs. Na's 1e⁻), leading to stronger metallic bonding.
- **Q:** Rank K, Li, and Na by decreasing metallic bond strength.
 - **A:** Li > Na > K (smallest to largest ionic radius).

2. VSEPR and Molecular Shape Questions

- **Q:** Predict the shape of SO₃²⁻.
 - **Steps:**
 - Lewis structure: S central, 6 + 3×6 + 2 = 26 e⁻; S has 1 lone pair + 3 bonding pairs.
 - Electron domains: 4 (3 bonding + 1 nonbonding) → tetrahedral domain geometry.
 - Molecular shape: Trigonal pyramidal.
- **Q:** Is XeF₄ polar?
 - **A:** No. Square planar shape (symmetrical) with identical F atoms → nonpolar.

3. Hybridization Questions

- **Q:** What is the hybridization of the central atom in CO₂?
 - **A:** Linear geometry → 2 electron domains → **sp** hybridization.
- **Q:** Identify the hybrid orbitals in NH₄⁺.
 - **A:** Tetrahedral shape → 4 electron domains → **sp³** hybridization.

V. Key Equations and Study Tips

1. Formulas to Remember

- **Electronegativity Difference (ΔEN):**
 - $\Delta EN > 2.0 \approx$ ionic bond; $0.4\text{--}2.0 \approx$ polar covalent; $<0.4 \approx$ nonpolar covalent.
- **Hybridization Shortcut:**
 - Electron domains = # hybrid orbitals (e.g., 5 domains = sp^3d).

VI. Intermolecular and Intramolecular Forces

- Intramolecular forces the forces that exist inside the molecules (only covalent bond)
- Intermolecular forces the forces of attraction between neighboring molecules
- 3 Intermolecular forces:

1. London Dispersion forces

- A force of attraction between induced dipoles, present in all molecular substances.
- Induced dipole: a dipole that is created by the presence of a neighboring dipole.
- Dispersion force: a force of attraction between induced dipoles.
- Larger atomic size, greater dispersion forces, higher boiling point.

2. Dipole-dipole forces

- Attractions between opposite charges of neighboring dipoles
- Polar molecules have permanent dipoles
- Whether a molecule is a permanent dipole depends on if they are polar bonds and the geometry of molecules.

3. Hydrogen Bonding

- The extremely strong dipole-dipole force between the H atom of one molecule and the N/O/F atoms of another molecule.